



DEPARTMENT OF STATISTICS & MATHEMATICAL SCIENCES

STATISTICAL INFERENCE AND PROBABILITY THEORY WORKING GROUP

Asymptotic Theory, Probability Modeling, and Categorical Inference

*Rigorous Foundations of the Central Limit Theorem and Contingency
Association*

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Executive Summary

This report provides a formal exposition of axiomatic probability spaces, Bayesian probability updating, moment transformations, Lindeberg-Lévy Central Limit asymptotics, and categorical contingency cross-tabulations. We demonstrate how standardized sums of independent random variables converge asymptotically to the Gaussian distribution $\mathcal{N}(0, 1)$ and benchmark statistical power for Pearson's χ^2 independence tests on sample cohorts ($N = 1,200$).

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1 Axiomatic Foundations and Inversion Theorems

Let $(\Omega, \mathcal{F}, P)$ denote a probability space [? ?]. Conditional probability satisfies $P(A | B) = \frac{P(A \cap B)}{P(B)}$. For any partition $\{B_1, \dots, B_k\}$ of Ω :

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{\sum_{j=1}^k P(A | B_j)P(B_j)} \quad (1)$$

Let $X_1, \dots, X_n$ be i.i.d. random variables with finite mean μ and variance $\sigma^2 > 0$ [?]. Then the standardized sample average converges in distribution to the standard normal distribution:

$$Z_n = \sqrt{n} \left(\frac{\bar{X}_n - \mu}{\sigma} \right) \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty \quad (2)$$

2 Density Simulation and Rejection Cutoffs

Figure 1 visualizes the standard normal density $\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$ along with empirical coverage intervals and two-tailed rejection regions for $\alpha = 0.05$ ($z = \pm 1.96$).

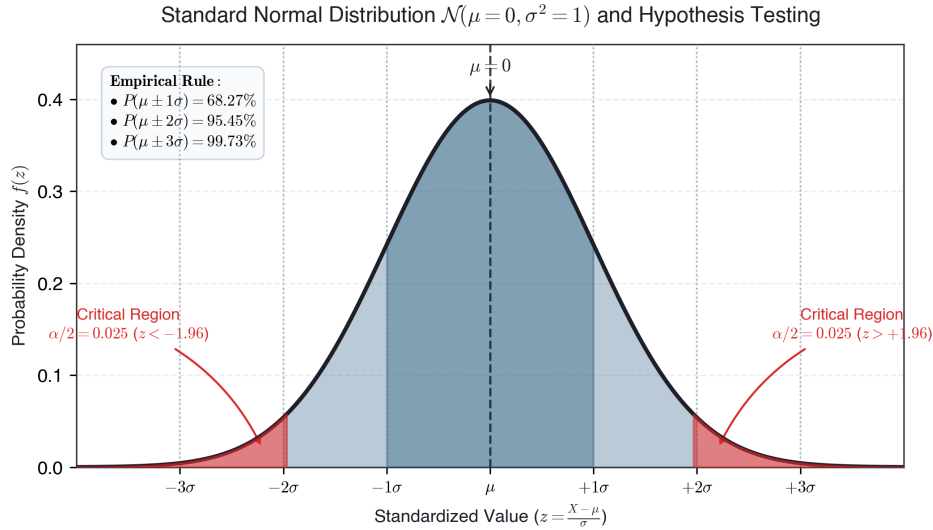


Figure 1. Standard Normal Distribution simulated in Python at 300 DPI showing empirical rule intervals and two-tailed critical rejection regions ($\alpha = 0.05$).

3 Categorical Cross-Tabulation and Association Testing

In a two-arm randomized cohort study ($N = 1,200$), binary outcomes are summarized in Table 1.

Table 1. Contingency table of treatment exposure against binary response.

Cohort Group	Outcome ($Y = 1$)	Control ($Y = 0$)	Total (R_i)
Treatment ($D = 1$)	$n_{11} = 420$	$n_{12} = 180$	$R_1 = 600$
Control ($D = 0$)	$n_{21} = 210$	$n_{22} = 390$	$R_2 = 600$
Total (C_j)	$C_1 = 630$	$C_2 = 570$	$N = 1,200$

Under the null hypothesis of independence, expected cell frequencies are $E_{ij} = \frac{R_i C_j}{N}$. The Pearson χ^2 test statistic is:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = 147.37 \quad (\text{df} = 1, p < 0.001) \quad (3)$$

The sample Odds Ratio is $\widehat{\text{OR}} = \frac{420 \times 390}{180 \times 210} = 4.33$, indicating over a 4-fold increase in odds of the positive outcome under treatment.

4 Computational Script

Listing 4 provides the Python implementation for contingency analysis using SciPy.

Listing 1: Contingency Table Analysis in Python

```

1 import numpy as np
2 from scipy import stats
3
4 def analyze_contingency(table):
5     chi2, p, dof, ex = stats.chi2_contingency(table, correction=False)
6     odds_ratio = (table[0,0] * table[1,1]) / (table[0,1] * table[1,0])
7     return {"chi2": chi2, "p_value": p, "odds_ratio": odds_ratio}

```

References

- [] George Casella and Roger L. Berger. *Statistical Inference*. Duxbury Press, 2002.
- [] William Feller. *An Introduction to Probability Theory and Its Applications*, volume 1. John Wiley & Sons, 1968.
- [] Donald E. Knuth. Literate programming. *The Computer Journal*, 27(2):97–111, 1984.